PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. A solid sphere of mass 'm' and radius 'r' is allowed to roll without slipping from the highest point of an inclined plane of length 'L' and makes an angle 30° with the horizontal. The speed of the particle at the bottom of the plane is v_1. If the angle of inclination is increased to 45° while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2. The ratio $v_1^2 : v_2^2$ is

(1) 1 : 3

(2) $1:\sqrt{2}$

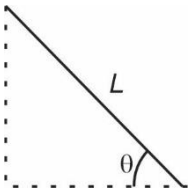
(3) 1 : 2

(4) $1:\sqrt{3}$

Answer (2)

Sol. Loss in P.E. = gain in K.E.

$$mgL\sin\theta = \frac{1}{2}\left(\frac{7}{5}\right)mv^2$$



$$\Rightarrow \quad v^2 \propto \sin\theta$$

$$\frac{v_1^2}{v_2^2} = \frac{\sin 30^\circ}{\sin 45^\circ} = \frac{1}{\sqrt{2}}$$

27. A gun fires a lead bullet of temperature 300 K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J, then the mass of the bullet is _____ grams.

(Latent heat of fusion of lead = 2.5×10^4 JKg^{-1} and specific heat capacity of lead 125 JKg^{-1} K^{-1})

(1) 20 (2) 5

(3) 10 (4) 15

Answer (3)

Sol. $Q = ms\Delta T + mL$

$625 = m \times 125 \times 300 + m \times 2.5 \times 10^4$

$625 = m\{3.75 + 2.5\} \times 10^4$

$$\Rightarrow \frac{625}{6.25} \times 10^{-4} \text{ kg} = 10\text{ g} = m$$

28. A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1. If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1, after one half-life of n_1?

(1) 1/8 (2) 1/4

(3) 4 (4) 8

Answer (2)

Sol. $\lambda_2 = 3\lambda \qquad \lambda_1 = \lambda$

$N = N_0 e^{-\lambda t}$

$N_1 = N_0 e^{-\lambda t}$

$N_2 = N_0 e^{-3\lambda t}$

$$\frac{N_2}{N_1} = e^{-2\lambda t}$$



for $t = \frac{\ln 2}{\lambda}$

$\Rightarrow \frac{N_2}{N_1} = e^{-2\lambda \frac{\ln 2}{\lambda}}$

$\frac{N_2}{N_1} = \frac{1}{4}$

29. The electric flux is $\phi = \alpha\sigma + \beta\lambda$

Where λ and σ are linear and surface charge density, respectively. $\left(\frac{\alpha}{\beta}\right)$ represents

(1) Area (2) Displacement

(3) Electric field (4) Charge

Answer (2)

Sol. $\alpha \equiv \frac{\phi}{\sigma}$

$\beta = \frac{\phi}{\lambda}$

$\frac{\alpha}{\beta} \equiv \frac{\lambda}{\sigma} \equiv$ displacement

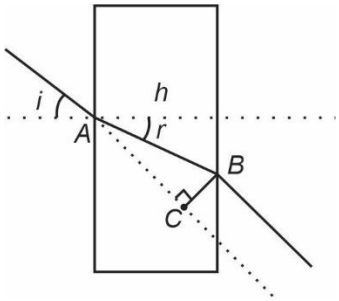
30. What is the lateral shift of a ray refracted through a parallel-sided glass slab of thickness '*h*' in terms of the angle of incidence '*i* ' and angle refraction '*r*', if the glass slab is placed in air medium?

(1) h (2) $\frac{h\sin(i-r)}{\cos r}$

(3) $\frac{h\tan(i-r)}{\tan r}$ (4) $\frac{h\cos(i-r)}{\sin r}$

Answer (2)

Sol.



$AB = h \sec r$

$BC = h \sec r \sin(i - r)$

$BC = \frac{h\sin(i-r)}{\cos r}$

31. Match the **List-I** with **List-II**

List-I		List-II	
A.	Pressure varies inversely with volume of an ideal gas.	I.	Adiabatic process
B.	Heat absorbed goes partly to increase internal energy and partly to do work.	II.	Isochoric process
C.	Heat is neither absorbed nor released by a system.	III.	Isothermal process
D.	No work is done on or by a gas.	IV.	Isobaric process

Choose the correct answer from the options given below:

(1) A-III, B-I, C-IV, D-II (2) A-I, B-IV, C-II, D-III

(3) A-III, B-IV, C-I, D-II (4) A-I, B-III, C-II, D-IV

Answer (3)

Sol. $PV = \mu RT$

(A) $P \propto \frac{1}{V} \Rightarrow$ isothermal

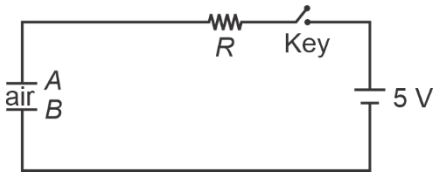
(B) $\Delta Q = \Delta V + W \Rightarrow$ isobaric

(C) $\Delta Q = 0 \Rightarrow$ adiabatic

(D) $W = 0 \Rightarrow$ isochoric



32. Identify the valid statements relevant to the given circuit at the instant when the key is closed.



A. There will be no current through resistor *R*.

B. There will be maximum current in the connecting wires.

C. Potential difference between the capacitor plates *A* and *B* is minimum.

D. Charge on the capacitor plates is minimum.

Choose the correct answer from the options given below:

(1) C, D only (2) A, B, D only

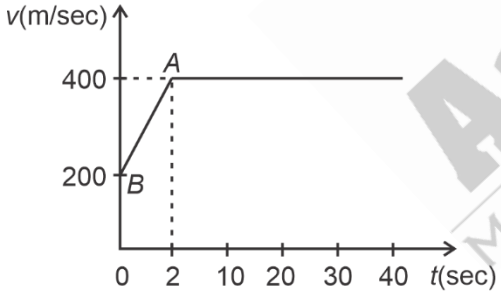
(3) A, C only (4) B, C, D only

Answer (4)

Sol. Capacitor behaves like closed circuit at $t = 0$ and charge is zero.

A *X*, B ✓, C ✓, D ✓

33. The motion of an airplane is represented by velocity-time graph as shown below. The distance covered by airplane in the first 30.5 second is _____ km.



(1) 12 (2) 3

(3) 6 (4) 9

Answer (1)

Sol. Distance = area under the graph

$d = 300 \times 2 + 400 \times 28.5$

$= 600 + 114000$

$= 12000$ m

34. Given below are two statements:

Statement I: The hot water flows faster than cold water.

Statement II: Soap water has higher surface tension as compared to fresh water.

In the light of above statements, choose the **correct** answer from the options given below

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

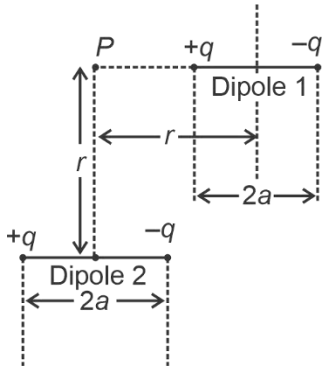
(3) Statement I is false but Statement II is true

(4) Statement I is true but Statement II is false

Answer (4)

Sol. Hot water flows faster because of less viscosity (✓) and soap water has less surface tension because bubbles are easily formed (*X*)

35. A point particle of charge *Q* is located at *P* along the axis of an electric dipole 1 at a distance *r* as shown in the figure. The point *P* is also on the equatorial plane of a second electric dipole 2 at a distance *r*. The dipoles are made of opposite charge *q* separated by a distance 2*a*. For the charge particle at *P* not to experience any net force, which of the following correctly describes the situation?



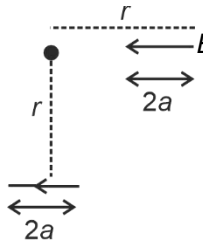
(1) $\frac{a}{r} \sim 3$ (2) $\frac{a}{r} \sim 0.5$

(3) $\frac{a}{r} \sim 10$ (4) $\frac{a}{r} \sim 20$

Answer (1)



Sol.



$$\Rightarrow \frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} = \frac{2a}{\left(a^2+r^2\right)^{3/2}}$$

$$\frac{4ar}{\left(r^2-a^2\right)^2} = \frac{2a}{\left(a^2+r^2\right)^{3/2}}$$

$$\left(r^2-a^2\right)^2 = 2r\left(a^2+r^2\right)^{3/2}$$

$$\left(1-\frac{a^2}{r^2}\right)^2 = 2\left(1+\frac{a^2}{r^2}\right)^{3/2}$$

$$\left(1-x^2\right)^2 = 2\left(1+x^2\right)^{3/2} \quad \left(x = \frac{a}{i}\right)$$

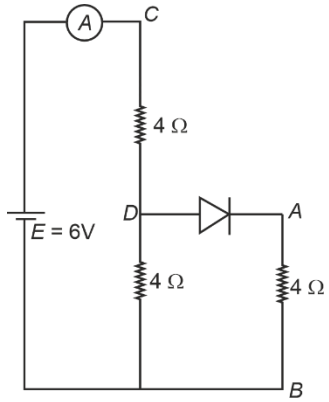
$$\frac{\left(1-x^2\right)^2}{\left(1+x^2\right)^{3/2}} = 2$$

Now for x = 3

We get $\frac{64}{10\sqrt{10}} \approx 2 \Rightarrow \frac{a}{r} \approx 3$

[But for $a > r$ point charge will between the dipole where $\vec{E} \neq 0$]

36. Refer to the circuit diagram given in the figure. which of the following observations are correct?



A. Total resistance of circuit is 6 Ω

B. Current in Ammeter is 1 A

C. Potential across *AB* is 4 volts.

D. Potential across *CD* is 4 volts

E. Total resistance of the circuit is 8 Ω.

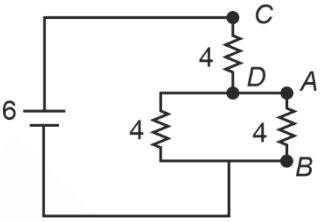
Choose the correct answer from the options given below:

(1) A, B and D Only (2) B, C and E Only

(3) A, B and C Only (4) A, C and D Only

Answer (1)

Sol.



R_{eq} (4 ↑↑ 4) + 4 = 6 Ω

$$i = \frac{6}{6}\text{A}$$

$$V_{CD} = 1 \times 4 = 4\text{V} \Rightarrow V_{AB} = 2\text{V}$$

(A, B, D)

37. Given a thin convex lens (refractive index μ_2), kept in a liquid (refractive index μ_1, $\mu_1 < \mu_2$) having radii of curvatures $|R_1|$ and $|R_2|$. Its second surface is silver polished. Where should an object be placed on the optic axis so that a real and inverted image is formed at the same place?

(1) $\frac{(\mu_2+\mu_1)|R_1|}{(\mu_2-\mu_1)}$

(2) $\frac{\mu_1|R_1|.|R_2|}{\mu_2(|R_1|+|R_2|)-\mu_1|R_1|}$

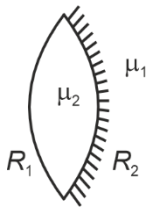
(3) $\frac{\mu_1|R_1|.|R_2|}{\mu_2(|R_1|+|R_2|)-\mu_1|R_2|}$

(4) $\frac{\mu_1|R_1|.|R_2|}{\mu_2(2|R_1|+|R_2|)-\mu_1\sqrt{|R_1|.|R_2|}}$



Answer (3)

Sol.



$$\frac{1}{f_l}=\left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

$$f_m=-\frac{R_2}{2}$$

$$p_{eq}=2p_l+pm$$

$$\frac{1}{|f|}=\frac{2(\mu_2-\mu_1)}{\mu_1}\left(\frac{1}{R_1}+\frac{1}{R_2}\right)+\frac{2}{R_2}$$

$$=\frac{2(\mu_2-\mu_1)(R_1+R_2)}{\mu_1(R_1R_2)}+\frac{2}{R_2}$$

$$\frac{1}{f}=\frac{2\mu_2R_1+2\mu_2R_2-2\mu_1R_1-2\mu_1R_2+2}{\mu_1R_1R_2}$$

$$\Rightarrow\quad f=\frac{\mu_1R_1R_2}{2\mu_2R_1+2\mu_2R_2-2\mu_1R_2}$$

Required distance = 2f.

38. A sub-atomic particle of mass 10^{-30} kg is moving with a velocity 2.21×10^6 m/s. Under the matter wave consideration, the particle will behave closely like _____.

($h = 6.63\times10^{-34}$ J.s)

(1) Gamma rays (2) X-rays

(3) Visible radiation (4) Infra-red radiation

Answer (2)

Sol. $$\lambda=\frac{6.63\times10^{-34}}{10^{-30}\times2.21\times10^6}$$

$\lambda = 3\times10^{-10}$

$\lambda = 3$ Å

$\Rightarrow$ X Ray

39. Regarding self-inductance:

A. The self-inductance of the coil depends on its geometry.

B. Self-inductance does not depend on the permeability of the medium.

C. Self-induced e.m.f. opposes any change in the current in a circuit.

D. Self-inductance is electromagnetic analogue of mass in mechanics.

E. Work needs to be done against self-induced e.m.f. in establishing the current.

Choose the correct answer from the options given below:

(1) A, C, D, E only (2) A, B, C, E only

(3) A, B, C, D only (4) B, C, D, E only

Answer (1)

Sol. (A) $L = \mu_r\mu_0n^2$ Al ✓

(B) $L = \mu_r\,\mu_v n^2$ Al ✗

(C) ✓

(D) ✓

(E) ✓

40. Consider a moving coil galvanometer (MCG):

A. The torsional constant in moving coil galvanometer has dimentions $[ML^2T^{-2}]$

B. Increasing the current sensitivity may not necessarily increase the voltage sensitivity.

C. If we increase number of turns (N) to its double (2N), then the voltage sensitivity doubles.

D. MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.

E. Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from the options given below:

(1) A, D Only (2) A, B Only

(3) B, D, E Only (4) A, B, E Only



Answer (2)

Sol. (A) $\tau = C\theta$

$C \equiv ML^2T^{-2}$ ✓

(B) $C\theta = iANB$

$\frac{\theta}{i} = \frac{ANB}{C}$ here $N\uparrow, \frac{\theta}{i}\uparrow$

$\frac{\theta}{v} = \frac{ANB}{CR}$ here $\frac{N}{R}$ cannot increase

✓

(C) X; Theoretical

(D) Shunt of law of resistance X

(E) X; explained in [B]

41. A light hollow cube of side length 10 cm and mass 10 g, is floating in water. It is pushed down and released to execute simple harmonic oscillations. The time period of oscillations is $y\pi \times 10^{-2}$ s, where the value of y is

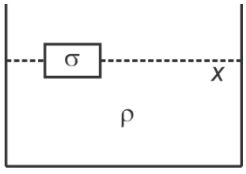
(Acceleration due to gravity, g = 10 m/s^2, density of water = 10^3 kg/m^3)

(1) 2 (2) 6

(3) 4 (4) 1

Answer (1)

Sol. Additional buoyant force



$g\rho\ a^2x = \sigma a^3 A$

$A = \frac{\rho}{\sigma}\frac{g}{a}x$

$T = 2\pi\sqrt{\frac{\sigma a}{\rho g}}$

Now, $\sigma = \frac{10\times10^{-3}}{10^{-3}} = 10$

$\Rightarrow T = 2\pi\sqrt{\frac{10\times0.1}{10^3\times10}}$

$= 2\pi\times10^{-2}$

42. The position of a particle moving on x-axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is

(1) L^2T^{-2} (2) L^3T^{-2}

(3) L (4) L^2

Answer (1)

Sol. $A \equiv L$

$B \equiv L$

$C \equiv LT^{-2}$

$D \equiv L$

$\Rightarrow \frac{ABC}{D} = \frac{L\times L\times LT^2}{L} = L^2T^{-2}$

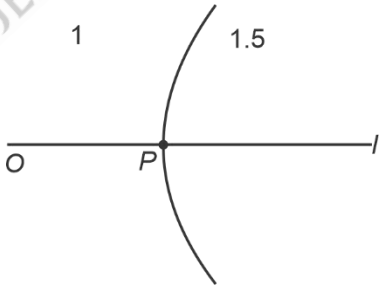
43. A spherical surface of radius of curvature R, separates air from glass (refractive index = 1.5). The centre of curvature is in the glass medium. A point object 'O' placed in air on the optic axis of the surface, so that its real image is formed at 'I' inside glass. The line OI intersects the spherical surface at P and $PO = PI$. The distance PO equals to

(1) $3R$ (2) $2R$

(3) $1.5R$ (4) $5R$

Answer (4)

Sol.



$PO = PI$

$\frac{1.5}{x} - \frac{1}{-x} = \frac{0.5}{R}$

$\frac{2.5}{x} = \frac{0.5}{R}$

$x = 5R$



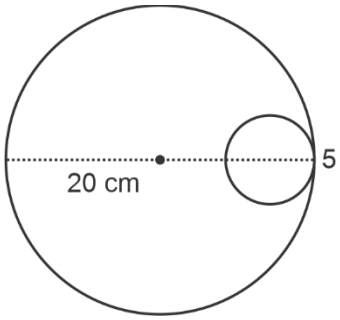
44. Consider a circular disc of radius 20 cm with centre located at the origin. A circular hole of radius 5 cm is cut from this disc in such a way that the edge of the hole touches the edge of the disc. The distance of centre of mass of residual or remaining disc from the origin will be

(1) 1.5 cm (2) 0.5 cm

(3) 1.0 cm (4) 2.0 cm

Answer (3)

Sol.



$M \propto A$

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$= \frac{A(0) + \left(-\frac{A}{16}\right)(15)}{A - \frac{A}{16}}$$

$$= \frac{-15}{16-1} = -1 \text{ cm}$$

45. The electric field of an electromagnetic wave in free space is

$\vec{E} = 57\cos\left[7.5\times10^6 t - 5\times10^{-3}(3x+4y)\right]$

$\left(4\hat{i} - 3\hat{j}\right)$N/C. The associated magnetic field in Tesla is

(1) $\vec{B} = -\frac{57}{3\times10^8}\cos\left[7.5\times10^6 t - 5\times10^{-3}(3x+4y)\right]\left(\hat{k}\right)$

(2) $\vec{B} = \frac{57}{3\times10^8}\cos\left[7.5\times10^6 t - 5\times10^{-3}(3x+4y)\right]\left(5\hat{k}\right)$

(3) $\vec{B} = -\frac{57}{3\times10^8}\cos\left[7.5\times10^6 t - 5\times10^{-3}(3x+4y)\right]\left(5\hat{k}\right)$

(4) $\vec{B} = \frac{57}{3\times10^8}\cos\left[7.5\times10^6 t - 5\times10^{-3}(3x+4y)\right]\left(\hat{k}\right)$

Answer (2)

Sol. $E = CB$

and $\vec{E}\times\vec{B} \uparrow\uparrow \vec{C}$

$$B = \frac{57\times5}{3\times10^8}$$

$3\hat{i} + 4\hat{j}$ is direction of propagation

As, $\vec{E}\times\vec{B} \uparrow\uparrow 3\hat{i} + 4\hat{j}$

$\Rightarrow \left(4\hat{i} - 3\hat{j}\right)\times\left(-\hat{k}\right) = 4\hat{j} + 3\hat{i}$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. A positive ion A and a negative ion B has charges 6.67×10^{-19} C and 9.6×10^{-10} C, and masses 19.2×10^{-27} kg and 9×10^{-27} kg respectively. At an instant, the ions are separated by a certain distance r. At that instant the ratio of the magnitudes of electrostatic force to gravitational force is $P \times 10^{-23}$, where the value of P is _____.

(Take $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ Nm^{2}C^{-1} and universal gravitational constant as 6.67×10^{-11} Nm^{2}kg^{-2})

Answer (None)

Sol. $F_e = \frac{k q_1 q_2}{r^2}$

$$F_g = \frac{G m_1 m_2}{r^2}$$

$$\frac{F_e}{F_g} = \frac{k q_1 q_2}{G m_1 m_2}$$

$$= \frac{9\times10^9 \times 6.67\times10^{-19} \times 9.6\times10^{-10}}{6.67\times10^{-11}\times19.2\times10^{-27}\times9\times10^{-27}}$$

$$= \frac{10^{-20}}{2\times10^{-65}}$$

***Answer does not match with description.**



47. An ideal gas initially at 0°C temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is 3/2, the change in temperature due to the thermodynamic process is _____ K.

Answer (273)

Sol. $TV^{\gamma-1} = \text{constant}$

$$273 \times V^{0.5} = T\left(\frac{V}{4}\right)^{0.5}$$

$$T = 546$$

$$\Delta T = 273 \text{ K}$$

48. A force $f = x^2y\hat{i} + y^2\hat{j}$ acts on a particle in a plane $x + y = 10$. The work done by this force during a displacement from (0, 0) to (4m, 2m) _____ Joule (round off to the nearest integer)

Answer (152)

Sol. $y = 10 - x$

$$w = \int_0^4 x^2(10-x)dx + \int_0^2 y^2 dy$$

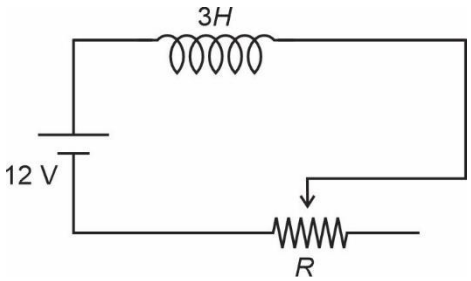
$$= \frac{10x^3}{3} - \frac{x^4}{4}\Big|_0^4 + \frac{y^3}{3}\Big|_0^2$$

$$= \frac{640}{3} - \frac{256}{4} + \frac{8}{3}$$

$$= 216 \times 64$$

$$= 152 \text{ J}$$

49.



In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of 8 A/s. At an instant when R is 12 Ω, the value of the current in the circuit will be _____ A.

Answer (1)

Sol. $\varepsilon - \frac{Ldi}{dt} = iR = 0$

$$12 - 3 \times 8 - i12 = 0$$

$$12 = i12$$

$$i = 1$$

50. Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A} = 2\hat{i} + 3n\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is _____.

Answer (3)

Sol. $\vec{A} \cdot \vec{B} = 0$ and $|\vec{A}| = |\vec{B}|$

$\Rightarrow$ $4 - 6n + 8p = 0$

$3n - 4p = 2$ …(i)

Also $4 + 9n^2 + 4 = 4 + 4 + 16p^2$

$3n = \pm 4p$ …(ii)

$\pm 4p - 4p = 2$

Taking –ve sign

$-8p = 2$

$$p = -\frac{1}{4}$$

$3n + 1 = 2$

$$n = \frac{1}{3}$$



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. 2.8×10^{-3} mol of CO_2 is left after removing 10^{21} molecules from its 'x' mg sample. The mass of CO_2 taken initially is

Given : $N_A = 6.02 \times 10^{23}$ mol^{-1}

(1) 98.3 mg (2) 48.2 mg

(3) 150.4 mg (4) 196.2 mg

Answer (4)

Sol. Moles of removed $CO_2 = \frac{10^{21}}{6.02 \times 10^{23}}$ mol

$= 1.66 \times 10^{-3}$ mol

mole of CO_2 left = 2.8×10^{-3} moles

total moles of CO_2 taken initially

$= (2.8 + 1.66) \times 10^{-3}$ mol

mass of CO_2 taken initially

$= 4.46 \times 10^{-3} \times 44$

$= 196.24 \times 10^{-3}$ g

= 196.24 mg

52. $FeO_4^{2-} \xrightarrow{+2.0\,V} Fe^{3+} \xrightarrow{0.8\,V} Fe^{2+} \xrightarrow{-0.5\,V} Fe^0$

In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $E^{\ominus}_{FeO_4^{2-}/Fe^{2+}}$ is

(1) 1.7 V (2) 2.1 V

(3) 1.4 V (4) 1.2 V

Answer (1)

Sol. $FeO_4^{2-} + 3e^- \rightarrow Fe^{3+}$; $E^\circ = 2.0$ eV

$Fe^{3+} + e^- \rightarrow Fe^{2+}$; $E^\circ = 0.8$ eV

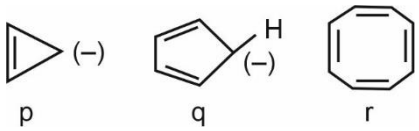
$FeO_4^{2-} + 4e^- \rightarrow Fe^{2+}$; $E^\circ = x$ (say)

$4 \times x = 3 \times 2 + 1 \times 0.8$

$x = \frac{6.8}{4}$ V

$= 1.7$ V

53. The correct stability order of the following species/molecules is



p q r

(1) q > p > r (2) q > r > p

(3) r > q > p (4) p > q > r

Answer (2)

Sol. q is aromatic

r is non-aromatic

p is antiaromatic

q > r > p (order of stability)

Aromatic > non-aromatic > antiaromatic

54. Ice at –5ºC is heated to become vapour with temperature of 110ºC at atmospheric pressure. The entropy change associated with this process can be obtained from

(1) $\int_{268K}^{273K} C_{p,m}dT + \frac{\Delta H_{m,fusion}}{T_f} + \frac{\Delta H_{m,vaporisation}}{T_b} + \int_{273K}^{373K} C_{p,m}dT + \int_{373K}^{383K} C_{p,m}dT$

(2) $\int_{268K}^{383K} C_p\,dT + \frac{\Delta H_{melting}}{273} + \frac{\Delta H_{boiling}}{373}$

(3) $\int_{268K}^{273K} \frac{C_{p,m}}{T}dT + \frac{\Delta H_{m,fusion}}{T_f} + \frac{\Delta H_{m,vaporisation}}{T_b} + \int_{273K}^{373K} \frac{C_{p,m}dT}{T} + \int_{373K}^{383K} \frac{C_{p,m}dT}{T}$

(4) $\int_{268K}^{383K} C_p dT + \frac{q_{rev}}{T}$

Answer (3)

